

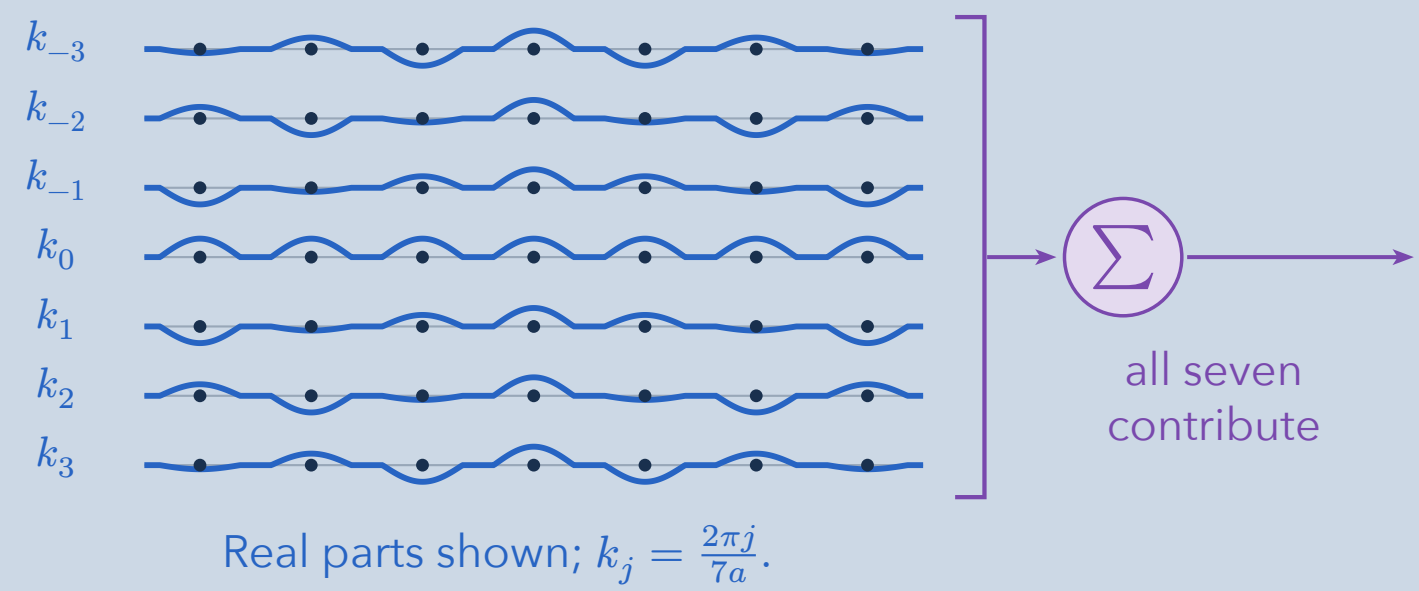
Wannierization & Wannier centers

From extended waves to local orbitals and their centers.

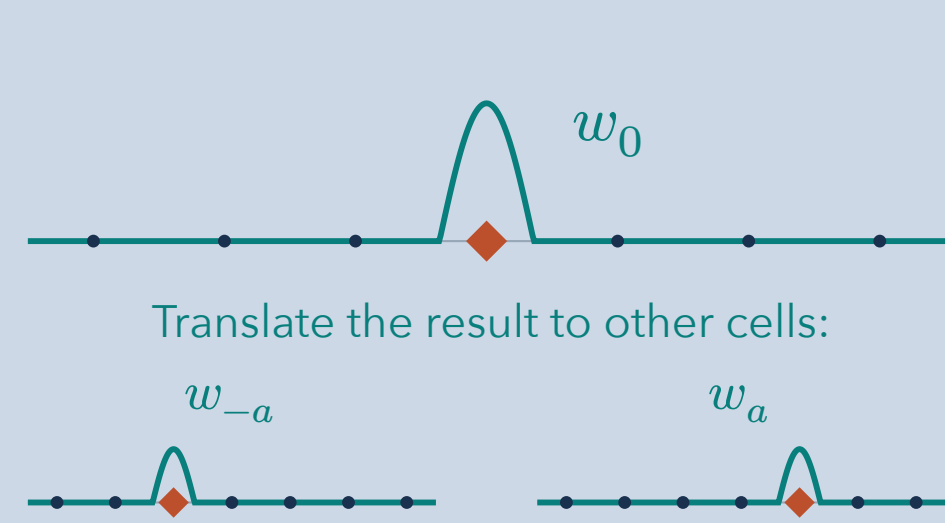
- nucleus
- ◆ Wannier center

1 CHOOSE PHASES, THEN ADD ALL THE WAVES

One-band toy model • seven states



One localized Wannier orbital



At the target cell: phases reinforce



At a neighboring cell: phases cancel



Each arrow is a unit complex phase. Add head to tail; the Fourier prefactor sets normalization.

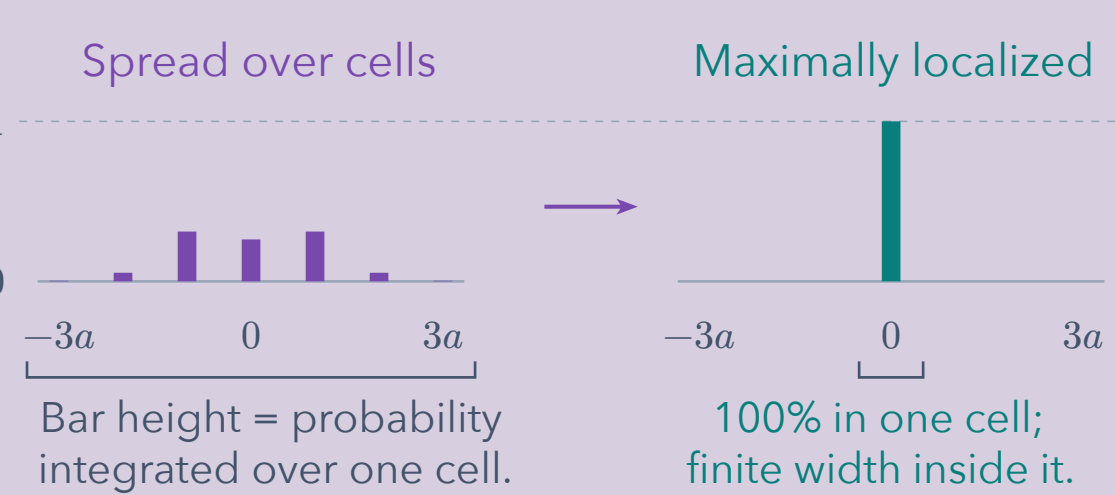
$$w_R(x) = \frac{1}{\sqrt{N_k}} \sum_k e^{-ikR} e^{i\varphi(k)} \psi_k(x)$$

$w_R(x)$: localized Wannier orbital for cell R .
 x : position coordinate in real space.
 R : lattice position of the chosen cell.
 a : lattice spacing; R is a multiple of a .
 $\sum_k$: sum over the sampled wavevectors.

$\psi_k(x)$: normalized, extended Bloch state.
 k : sampled crystal wavevector.
 N_k : number of sampled k points (here 7).
 $\varphi(k)$: chosen Bloch phase (here 0).
 $\frac{1}{\sqrt{N_k}}$: normalization for orthonormal states.

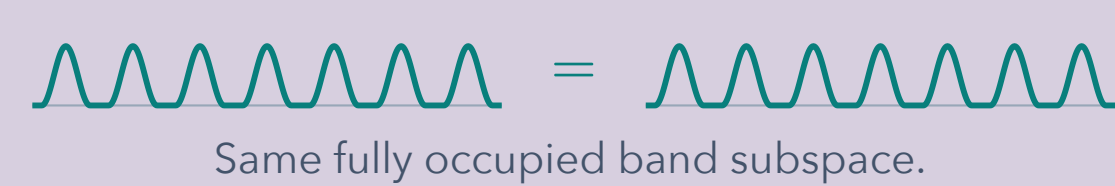
2 CHOOSE LOCALIZATION

Same physical state; change the basis.



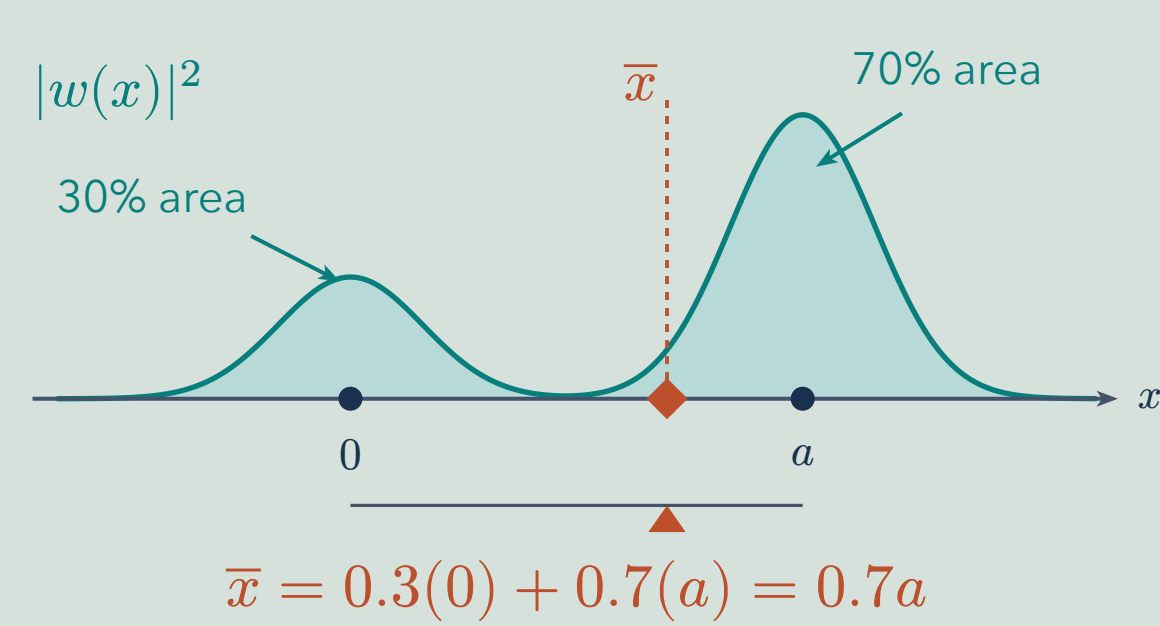
$$\text{Minimize } \Omega = \sum_n \left(\langle r^2 \rangle_n - |\bar{r}_n|^2 \right).$$

Total occupied density: unchanged



3 MEASURE THE CENTER

Probability density $\rightarrow$ mean position



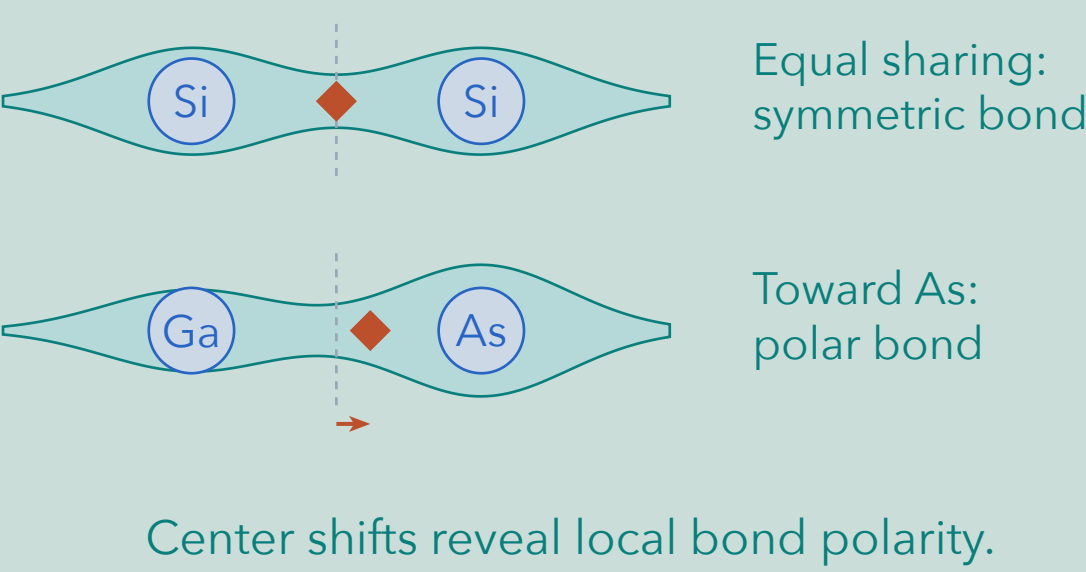
The mean need not coincide with a peak or nucleus.

$$\bar{r}_n = \int \mathbf{r} |w_{n0}(\mathbf{r})|^2 d^3r$$

4 USE THE LOCAL PICTURE

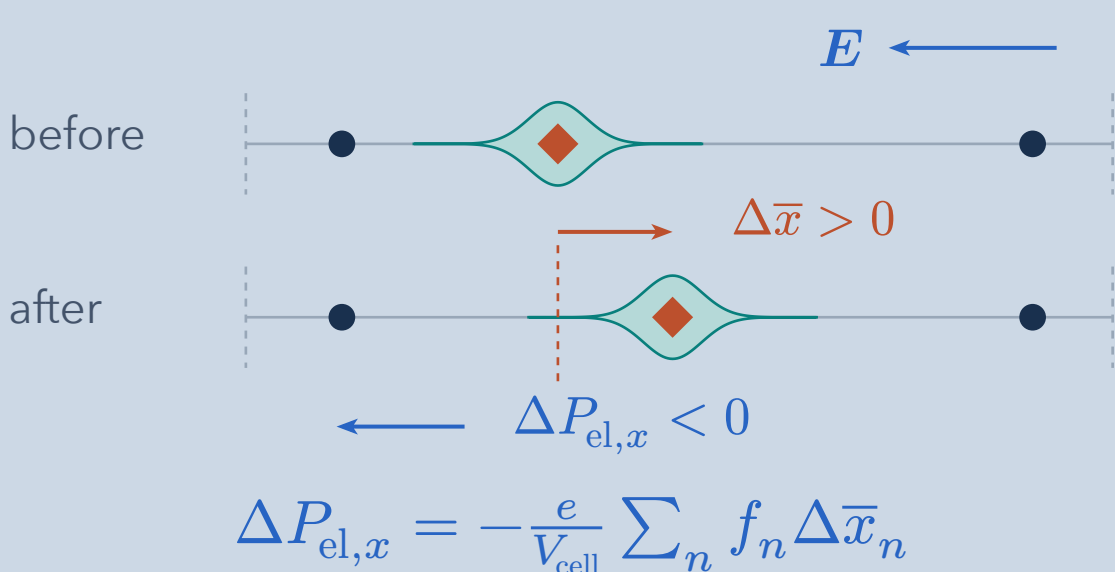
A READ CHEMICAL BONDS

Valence orbitals; density shapes schematic.



B FOLLOW A PHYSICAL RESPONSE

Apply an electric field; keep ions fixed.



$e > 0$; f_n is the occupation (2 for a pair). Fixed cell volume V_{cell} ; remain insulating.

Multiband: unitary U

$$w_{n\mathbf{R}} = \frac{1}{\sqrt{N_k}} \sum_{\mathbf{k}} e^{-i\mathbf{k} \cdot \mathbf{R}} \sum_m U_{mn}(\mathbf{k}) \psi_{m\mathbf{k}}$$

Individual centers depend on gauge U . Track them across cell boundaries; add ions for total polarization.